Cantera vs NN | Test IC 39 | NRMSE [%]

T0 = 1250.1 K P0 = 2.297 atm phi0 = 0.646 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	28.8	2.32	9.08	12.5	16.8	370	24.1	6.81	9.36	N/A	Complete
2 P-only DeepONet10	**10k**	0.755	1.15	3.32	16.4	0.341	21.7	0.661	2.89	2.68	N/A	Complete
2 P-only DeepONet10	**long**	0.0471	0.0723	0.267	0.179	0.17	1.8	0.0287	0.403	0.546	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	1.07	1.13	3.08	18.2	0.374	22.3	0.699	2.55	1.99	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	0.122	0.178	0.985	1.16	0.19	1.57	0.155	0.33	0.445	N/A	Complete
4 T/P DeepONet10	**10k**	7.79	1.03	3.04	9.45	3.31	60.4	4.88	6.71	8.88	N/A	Complete
4 T/P DeepONet10	**long**	4.79	0.203	1.62	1.18	2.68	51.9	3.4	1.79	2.53	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	6.96	0.633	2.49	7.87	3.52	60.1	4.36	7.28	9.86	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	65.1	1.85	50.5	6.09	29.8	634	41.9	7.95	10.8	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	7.83	0.959	3.45	7.86	3.48	64.4	4.98	5.54	6.34	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	0.0837	0.0861	0.123	0.248	0.0971	0.736	0.0874	0.149	0.175	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	6.74	0.59	3.05	4.33	3.34	63	4.46	5.99	7.74	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.0724	0.244	0.405	1.66	0.453	7.13	0.176	0.198	0.881	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.